Data valuation: which data helps, and for what?

- Data valuation measures a contribution to a specified objective.
- The learning procedure and evaluation population determine what the score means.
- Today: influence functions and Shapley background, then KAIROS in depth.

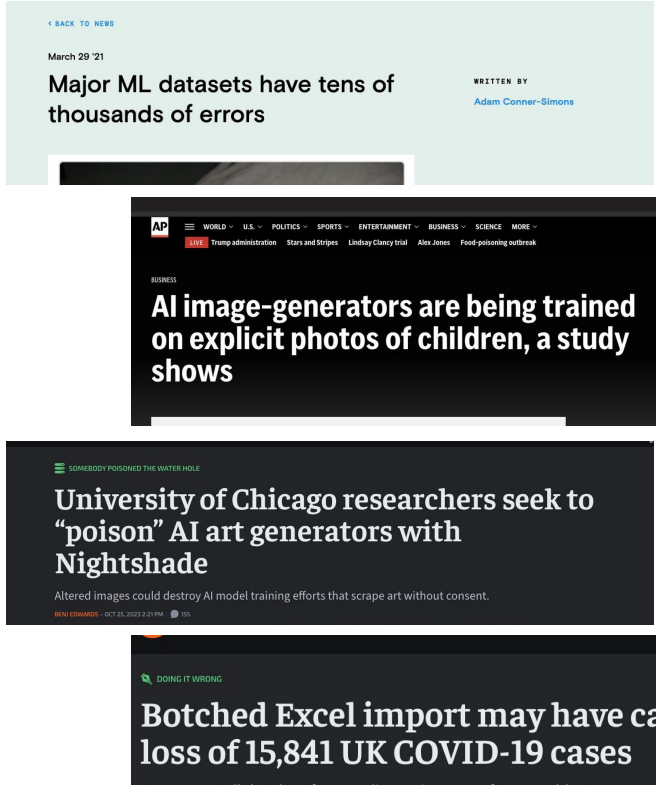
The case for trustworthy AI

- AI increasingly shapes decisions that affect people's lives: who gets considered for a job, who receives credit, and who gets access to health care.
- Greater autonomy raises the stakes. As AI systems act with less human oversight, there are fewer opportunities to catch mistakes before they cause harm.
- We have already seen AI systems fail in consequential ways: they have favored one group over another in hiring, contributed to wrongful arrests through facial-recognition errors, fabricated legal citations, reproduced sensitive training text, and become less safe after fine-tuning.
- This motivates trustworthy AI: systems that are accurate, reliable, and robust in their intended use, while also being fair, privacy-preserving, safe, secure, transparent, and accountable.



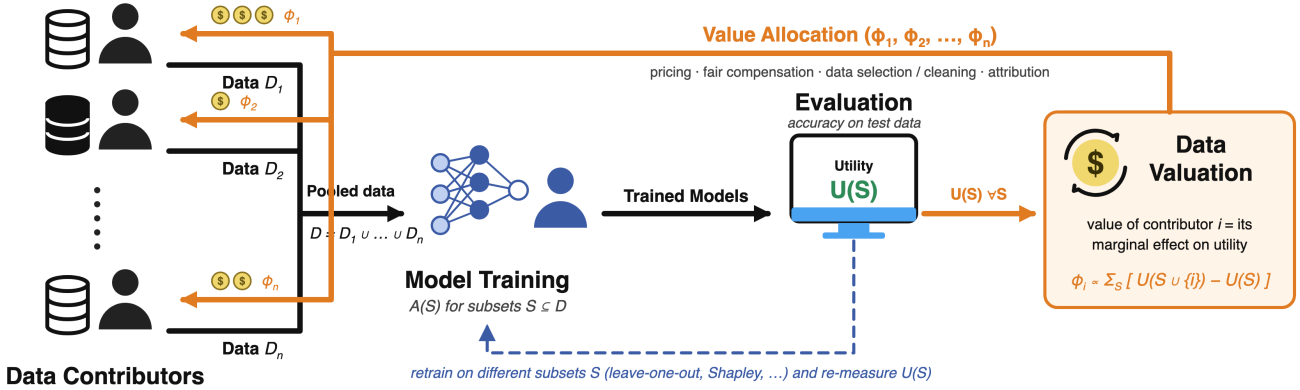
Data is a powerful point of intervention

- Data shapes what models learn: the capabilities they develop, the patterns they rely on, and the mistakes they make.
- Data problems can undermine learning and evaluation: biased samples, class imbalance, missing or corrupted values, incorrect or inconsistent labels, inconsistent formats or units, duplicates, outdated records, train–test leakage, and poisoned or harmful content.
- My research focuses on understanding how data shapes learning and developing principled methods to improve the data that learning systems rely on.



Data valuation

Data valuation assigns a score to a data contribution based on how much it helps or hurts a chosen objective.



- The unit can be one example, a batch, or a whole dataset.
- The objective sets the score: model performance or a data-level measure.
- A contribution can be negative. The same data can have different value for different tasks.

What data valuation enables

Data selection

Choose a compact coreset for efficient training.

Data reweighting

Use data values to adjust how strongly each example contributes to training.

Data minimization

Retain less data while limiting loss in task performance.

Data acquisition

Choose which data to collect or buy next.

Active learning

Spend the labeling budget on examples expected to help most.

Domain adaptation

Prioritize source examples that help on the target domain.

Data pricing

Use a dataset's contribution to help determine its price.

Explanation and debugging

Trace predictions and failures back to influential training examples.

Data cleaning

Find errors and possible poisoning for inspection and repair.

From model-dependent value to KAIROS

- Explain what a local influence score measures.
- Explain how Shapley allocates contribution across coalitions.
- Read KAIROS as a change of objective: value data without fixing a downstream learner.

Influence functions: trace a prediction back to training data

Pang Wei Koh & Percy Liang · ICML 2017

Understanding Black-box Predictions via Influence Functions

Pang Wei Koh [1] **Percy Liang** [1]

- Problem: refitting after every training-data change is expensive.
- Koh and Liang estimate the effect of infinitesimally upweighting one training point.

A training perturbation changes parameters, then test loss

Koh & Liang · original parameter and loss influence equations

z: training point · ztest: test point · θ̂: fitted weights · ε: added weight · L: loss · Hθ̂: risk Hessian. Classical assumptions: twice-differentiable risk and positive-definite Hθ̂.

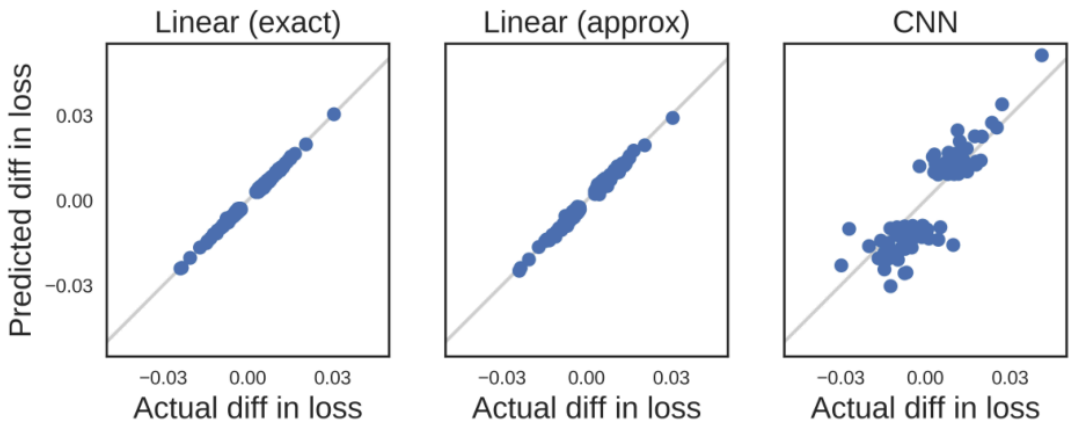
$$\mathcal{I}_{\text{up,params}}(z) \overset{\text{def}}{=} \left.\frac{d\hat{\theta}_{\epsilon,z}}{d\epsilon}\right|_{\epsilon=0} = -H_{\hat{\theta}}^{-1}\,\nabla_\theta L(z,\hat{\theta}), \qquad (1)$$

$$\begin{aligned}\mathcal{I}_{\text{up,loss}}(z, z_{\text{test}}) &\overset{\text{def}}{=} \left.\frac{dL(z_{\text{test}},\hat{\theta}_{\epsilon,z})}{d\epsilon}\right|_{\epsilon=0} \qquad (2)\\ &= \nabla_\theta L(z_{\text{test}},\hat{\theta})^\top \left.\frac{d\hat{\theta}_{\epsilon,z}}{d\epsilon}\right|_{\epsilon=0}\\ &= -\nabla_\theta L(z_{\text{test}},\hat{\theta})^\top H_{\hat{\theta}}^{-1}\nabla_\theta L(z,\hat{\theta}).\end{aligned}$$

- The training gradient gives a direction; inverse curvature gives the fitted model's local response.
- The test-loss gradient converts that parameter response into a loss change.
- Positive loss influence means harmful upweighting; finite deletion is a first-order approximation.

Influence evidence: compare with actual removal and retraining

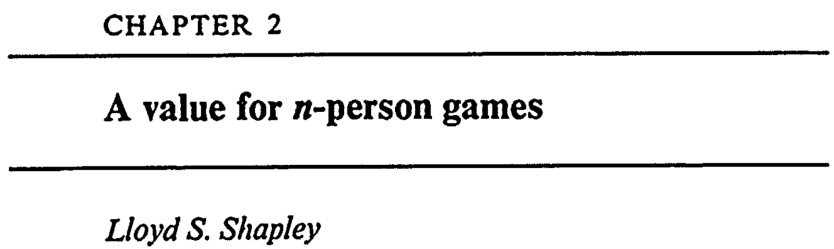
Koh & Liang · ICML 2017



- MNIST: compare predicted and actual removal effects for one wrongly classified query.
- Agreement in these models supports a local approximation—not universal fidelity.

Shapley: allocate the value created by a coalition

Lloyd S. Shapley · 1953 chapter · Reprinted in Roth, 1988



CHAPTER 2

A value for *n*-person games

Lloyd S. Shapley

- Problem: one full-dataset removal measures only one contribution context.
- Shapley allocates value by averaging marginal contribution over player arrival orders.

The formula averages marginal value across contexts

Shapley · original weights and value formula

N: n relevant players · U: player universe · S: coalition, s = |S| · v(S): coalition value · $\phi_i[v]$: credit · $\gamma_n(s)$: arrival-order weight

$$\gamma_n(s) = (s-1)!(n-s)!/n!, \tag{12}$$

$$\phi_i[v] = \sum_{S \subseteq N} \gamma_n(s)[v(S) - v(S-(i))] \qquad (\text{all } i \in U), \tag{13}$$

- For each player, average its added value across all possible predecessor coalitions.
- Symmetry, dummy-player behavior and additivity constrain the allocation; efficiency fixes its total.

Data Shapley: make training data the players

Amirata Ghorbani & James Zou · ICML 2019

Data Shapley: Equitable Valuation of Data for Machine Learning

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Amirata Ghorbani [1] **James Zou** [2]

- Fix a learning algorithm and a performance measure; train and evaluate on each data subset.
- The contribution is relative to that learner, metric, population and surrounding dataset.

Truncated Monte Carlo approximates coalition credit

Data Shapley · original truncated Monte Carlo algorithm

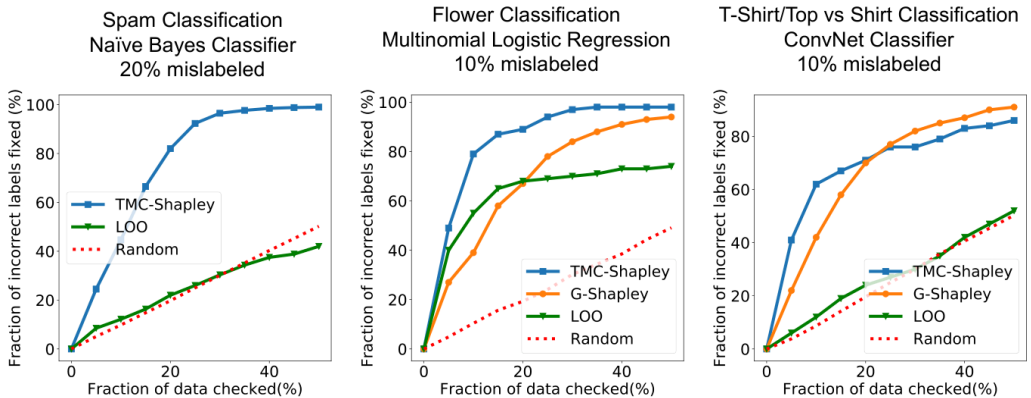
Algorithm 1 Truncated Monte Carlo Shapley

Input: Train data $D = \{1, \ldots, n\}$, learning algorithm $\mathcal{A}$, performance score V
Output: Shapley value of training points: $\phi_1, \ldots, \phi_n$
Initialize $\phi_i = 0$ for $i = 1, \ldots, n$ and $t = 0$
while Convergence criteria not met **do**
  $t \leftarrow t + 1$
  π^t: Random permutation of train data points
  $v_0^t \leftarrow V(\emptyset, \mathcal{A})$
  for $j \in \{1, \ldots, n\}$ **do**
    if $|V(D) - v_{j-1}^t| <$ Performance Tolerance **then**
      $v_j^t = v_{j-1}^t$
    else
      $v_j^t \leftarrow V(\{\pi^t[1], \ldots, \pi^t[j]\}, \mathcal{A})$
    end if
    $\phi_{\pi^t[j]} \leftarrow \frac{t-1}{t}\phi_{\pi^{t-1}[j]} + \frac{1}{t}(v_j^t - v_{j-1}^t)$
  end for
end for

- Sample a permutation; measure each arriving datum's change in performance.
- Average these increments across permutations; truncate when performance is within tolerance of the full set.

Data Shapley evidence: prioritize a label audit

Ghorbani & Zou · ICML 2019



- Inspect training labels in the order suggested by each method.
- The curve measures recovered corruption versus the fraction checked.
- This supports prioritizing an audit, followed by verification.

KAIROS changes the objective before computing value

- Influence and Data Shapley inherit the selected learner's loss or predictive performance.
- KAIROS values the data through discrepancy from a reference distribution.
- The reference and discrepancy now define what "valuable" means.

Model-agnostic data valuation

KAIROS: Scalable Model-Agnostic Data Valuation

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NeurIPS 2025



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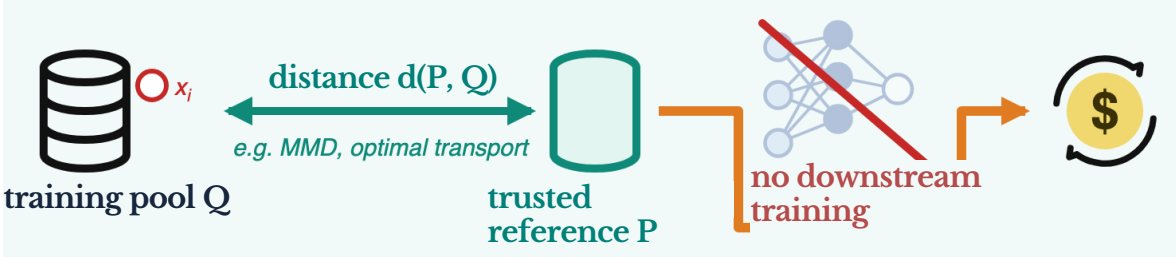
* Equal contribution

THE QUESTION

How can we value data without committing to a particular learning algorithm?

An emerging research direction: data-centric, model-agnostic valuation.

A data-level objective for valuation



Instead of training a model on Q and evaluating it on a clean, representative validation set P, we directly measure the distributional distance d(P, Q).

We value each training point by how it changes the distance between Q and P.

For the same model trained on Q, prior work establishes bounds for many distances, under suitable assumptions:

Target loss ≤ source loss + $C \cdot d(P, Q)$ + additional terms.

From a dataset discrepancy to a point's value



Before removal: Q and fixed P

remove z_i



After removal: Q without the point, same P

$$v_{LOO}(z_i) = d(P, Q_{-i}) - d(P, Q)$$

- Keep the reference P fixed and **remove one point** from the pool Q.
- If discrepancy increases, the point was **helping**. Its value is positive.
- If discrepancy decreases, the point was **hurting**. Its value is negative.
- Exact leave-one-out can require a new discrepancy computation **for every point**.

Which distance?

A value is a change in d(P, Q). Which distance?

- For **disjoint empirical supports**, KL can be infinite and total variation maximal, even when the samples are nearby.
- **Wasserstein and kernel MMD** use relationships between sample locations to capture more informative differences.

Wasserstein-1 and MMD share a test-function view: integral probability metrics.

Both are integral probability metrics

An IPM chooses a **critic** (or test function) whose average score differs most between P and Q.

$$IPM_F(P,Q) = \sup_{f \in F} \left| E_P[f] - E_Q[f] \right|$$

F: allowed critics, $f: X \to \mathbb{R}$

Wasserstein-1 the transport distance with an IPM form

$$W_1(P,Q) = \sup_{Lip(f) \le 1} \left| E_P[f] - E_Q[f] \right|$$

1-Lipschitz critics: scores change no faster than distances between inputs.

Classical exact OT: $O(n^3 \log n)$
+ $O(dn^2)$ to compute pairwise distances.

MMD

$$MMD(P,Q) = \sup_{\|f\|_H \le 1} \left| E_P[f] - E_Q[f] \right|$$

RKHS unit ball: kernel-defined critics with norm at most one.

Gaussian MMD: $O(dn^2)$
Closed-form pairwise kernel sums.

n = number of data points in each dataset; d = data dimension (number of features).

RKHS and MMD: three simple ideas

1. **A kernel** k(x,y) measures similarity.

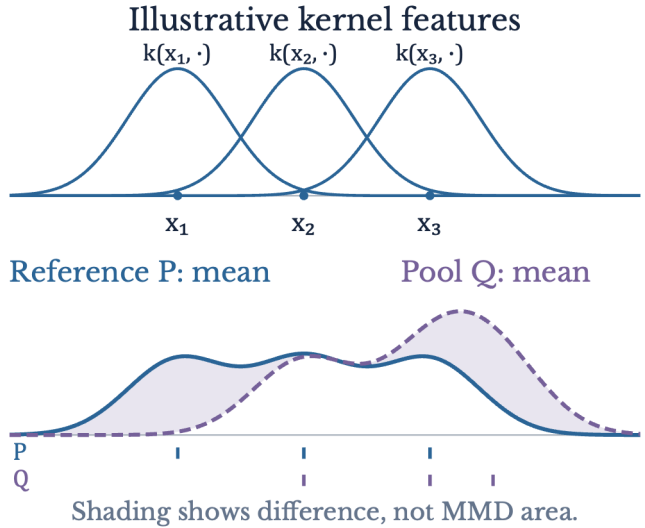
Point feature: $\varphi(x) = k(x,\cdot)$

The RKHS contains weighted sums of these feature functions.

$$\langle k(x,\cdot), k(y,\cdot)\rangle_H = k(x,y)$$

2. **Dataset becomes an average.** Average its feature functions to get its mean embedding.

3. **MMD compares the averages.** Small MMD means the pool looks like the reference.



$$\mathrm{MMD}_k(P,Q) = \|\mu_P - \mu_Q\|_H$$

RKHS norm

$$\mathrm{MMD}_k{}^2(P,Q) = \underbrace{E_{X,X'\sim P}\,k(X,X')}_{\text{within P}} + \underbrace{E_{Y,Y'\sim Q}\,k(Y,Y')}_{\text{within Q}} - 2\underbrace{E_{X\sim P,Y\sim Q}\,k(X,Y)}_{\text{across P and Q}}$$

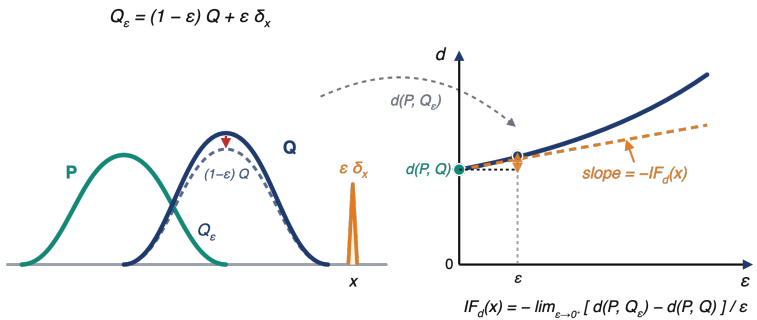
Closed form from **pairwise similarities**, without explicit feature vectors.

First-order influence of a training point

Approximating leave-one-out

Removing one of many points slightly changes the empirical distribution. We estimate the change in distance using its local sensitivity to that point's weight.

$$\mathrm{IF}_d(x; P, Q) = -\frac{d\big(P, (1-\varepsilon)Q + \varepsilon\delta_x\big) - d(P, Q)}{\varepsilon}\Big|_{\varepsilon \to 0^+}, \tag{1}$$



Reweight x and renormalize the other points. Features and labels stay fixed.

What does the sign mean?

Positive influence: more weight on x reduces the discrepancy.

For n > 1 equally weighted points. The removal approximation assumes differentiability across ε = 0.

Zhu, Prashant, Cloninger, Salimi. KAIROS: Scalable Model-Agnostic Data Valuation. NeurIPS 2025.

IPM sensitivity: two terms

Expanding the influence definition gives the following decomposition.

$$IF_d(x;P,Q) = \underbrace{\left.\frac{\mathbb{E}_Q[f_\varepsilon^\star - f^\star] - \mathbb{E}_P[f_\varepsilon^\star - f^\star]}{\varepsilon}\right|_{\varepsilon\to 0^+}}_{\text{(i) critic-shift term}} + \underbrace{\left(f^\star(x) - \mathbb{E}_Q[f^\star]\right)}_{\text{(ii) point-gap term}}. \tag{3}$$

f*: optimal critic before the nudge. f*ε: optimal critic after the nudge.

Critic-shift term

Loss in the original score gap per unit nudge.

Point-gap term

How much x's score exceeds the pool's average, using the original critic.

Can we compute influence using the original critic?

Reweighting can change which critic best distinguishes P and Q. We must establish when the original critic's point-gap gives the correct influence.

Zhu, Prashant, Cloninger, Salimi. KAIROS: Scalable Model-Agnostic Data Valuation. NeurIPS 2025.

Closed-form MMD influence and its cost

The MMD derivative is just kernel similarity.

Proposition 1. *The influence function for* MMD *as the distance metric is, up to additive and positive multiplicative constants, given by*

$$\mathrm{IF}_{\mathrm{MMD}}(x; P, Q) = \mathbb{E}_{x' \sim P}[k(x', x)] - \mathbb{E}_{x' \sim Q}[k(x', x)]. \tag{6}$$

- Higher similarity to the trusted reference raises the score.
- Higher similarity to the current pool lowers the score.

Closed-form scores: $O(n + m)$ per point; $O(n^2 + nm)$ for all n points.

Recomputing MMD for all n removals: $O(n^3 + n^2m)$ total.

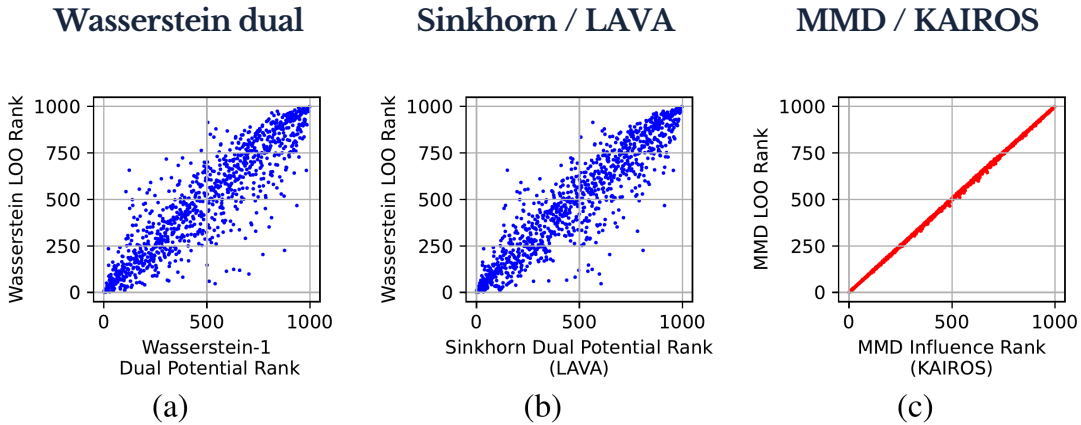
n training points, m reference points. Kernel evaluations; a dense Gaussian kernel adds a factor d.

Zhu, Prashant, Cloninger, Salimi. KAIROS, NeurIPS 2025, Eq. (7) and §3.4. Recompute cost derived from pairwise sums.

Do the scores predict the effect of removing data?

Comparison with prior work: LAVA (Just et al., ICLR 2023).

LAVA values data using sensitivity of a class-wise Wasserstein distance. Here we compare its transport-based scores with KAIROS and actual point removal.



One dot per training point. x: rank by score. y: rank by leave-one-out removal.

In this experiment, top-100 overlap with leave-one-out is **99% for KAIROS** and 60% for LAVA.

Just et al. LAVA: Data Valuation without Pre-Specified Learning Algorithms. ICLR 2023.
Zhu, Prashant, Cloninger, Salimi. KAIROS: Scalable Model-Agnostic Data Valuation. NeurIPS 2025.

Properties of KAIROS valuation

- **Symmetry**: equivalent effects on the discrepancy give equal scores.
- **Density separation**: under the proposition's assumptions and a suitable Gaussian bandwidth, scores separate specified density regions.
- **Labels**: a conditional discrepancy compares label behavior between the pool and reference.
- **Streaming**: reuse existing kernel sums and update them for new data.

Proposition 2 (Symmetry). *Let D^{train} and D^{val} be finite samples from distributions Q and P, respectively. If for all subsets $S \subseteq D^{train} \setminus \{x_i, x_j\}$,*

$$\widehat{\mathrm{MMD}}(D^{val}, S \cup \{x_i\}) - \widehat{\mathrm{MMD}}(D^{val}, S) = \widehat{\mathrm{MMD}}(D^{val}, S \cup \{x_j\}) - \widehat{\mathrm{MMD}}(D^{val}, S)$$

then $\widehat{\mathrm{IF}}(x_i) = \widehat{\mathrm{IF}}(x_j)$.

Proposition 3 (Density Separation). *Let P and Q be two probability distributions on $\mathcal{X} \subseteq \mathbb{R}^n$. For any $\epsilon > 0$ and $r > 0$, there exists a Gaussian isotropic kernel k such that for* $\mathrm{IF}(x) = \mathbb{E}_{x' \sim P}[k(x, x')] - \mathbb{E}_{x' \sim Q}[k(x, x')]$, *the following holds*

$$\begin{cases} \mathrm{IF}(x) > 0, & \text{if } P(x') - Q(x') \geq \epsilon \, \forall \, x' \in B(x, r), \\ \mathrm{IF}(x) < 0, & \text{if } P(x') - Q(x') \leq -\epsilon \, \forall \, x' \in B(x, r), \end{cases}$$

where $B(x, r) = \{x' : \|x' - x\| < r\}$.

Proposition 4. *The influence function for* E-MCMD *as the distance metric is, up to additive and positive multiplicative constants, given by*

$$\mathrm{IF}_{\text{E-MCMD}}(x, y; P, Q) = -\|\mu^P_{Y|X}(x) - \phi(y)\|_{\mathcal{H}_\mathcal{Y}} \tag{8}$$

For existing points (x_i, y_i) where $i \leq n_t$, the terms $\mathcal{B}_i$ and $\mathcal{R}_i$ remain unchanged. The training kernel average $\mathcal{A}_i$ can be efficiently updated as:

$$\mathcal{A}_i^{(t+1)} = \frac{1}{n_{t+1} - 1} \left((n_t - 1) \cdot \mathcal{A}_i^{(t)} + \sum_{j=1}^{m} k(x^{\text{train}}_{n_t + j}, x_i) \right)$$

The kernel form supports interpretable scores and efficient updates.

A bound linking discrepancy to transfer loss

Feature and label discrepancies control a loss-gap bound under stated assumptions.

Theorem 1 (Bounding transfer loss (simplified)). *Let $(\mathcal{X}, d_{\mathcal{X}})$ and $(\mathcal{Y}, d_{\mathcal{Y}})$ be compact metric spaces and $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$. Let $L : \mathcal{Z} \to \mathbb{R}$ be a continuous loss function. Then, for some constant c,*

$$\mathbb{E}_{(x,y)\sim Q}[L(x,y)] \leq \mathbb{E}_{(x,y)\sim P}[L(x,y)] + c\Big(\mathrm{MMD}(P_X, Q_X) + \mathbb{E}_{x\sim Q_X}\left[\mathrm{MCMD}_{P,Q}(x)\right]\Big).$$

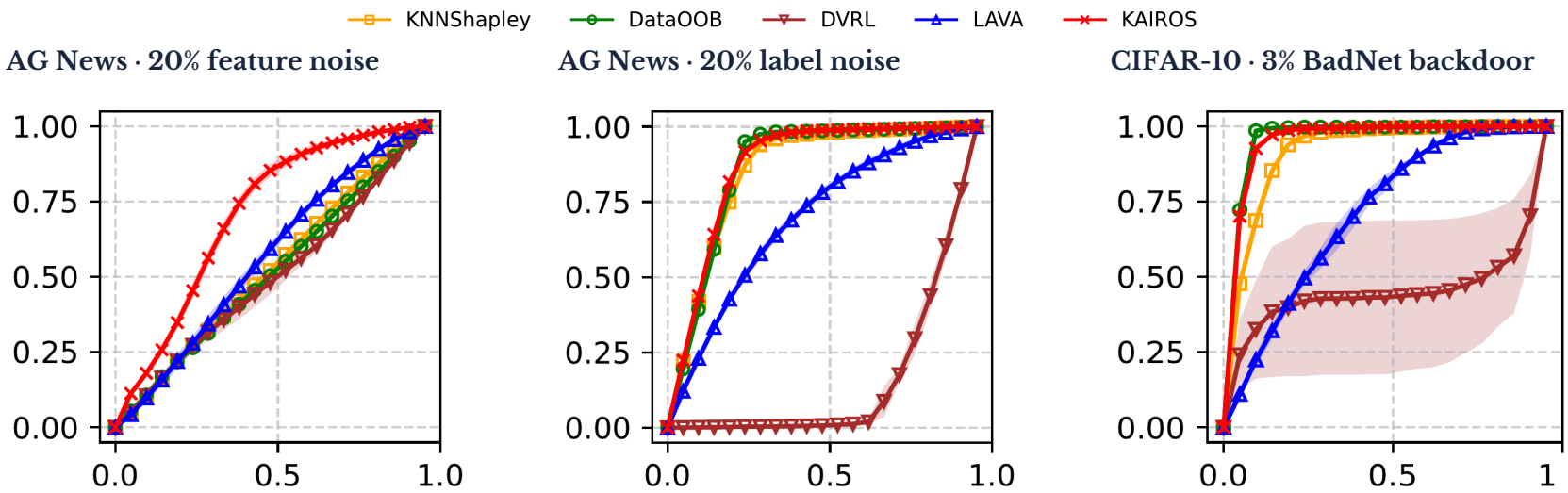
The theorem bounds loss using **feature mismatch and label mismatch**. Reducing those discrepancies tightens the bound's discrepancy term.

Tightens a bound; not a guarantee of better test accuracy.

Zhu, Prashant, Cloninger, Salimi. KAIROS: Scalable Model-Agnostic Data Valuation. NeurIPS 2025.

Finding corrupted examples early

Inspect from the lowest score upward. How much of the corruption is found?



x: fraction inspected. y: fraction of corrupted points found.

BadNets: trigger-marked images with wrong labels (3% poisoned).

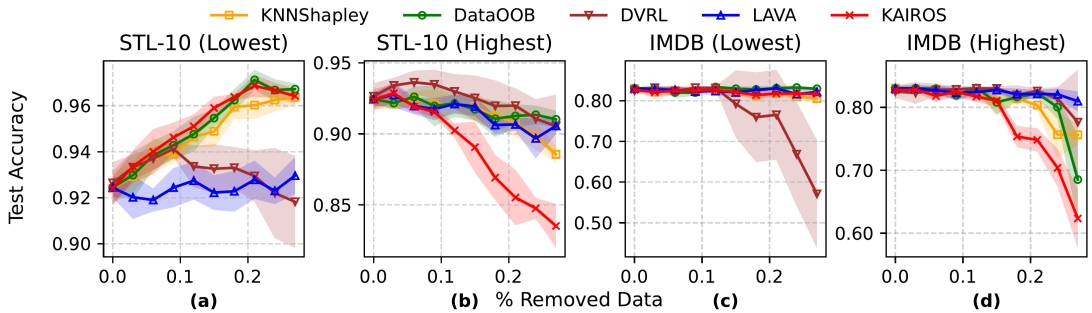
Across eight feature/label-noise settings, KAIROS ranks **first in six and second in two**. The backdoor panel is a separate test.

Zhu, Prashant, Cloninger, Salimi. KAIROS: Scalable Model-Agnostic Data Valuation. NeurIPS 2025.

Testing the ranking and scaling to ImageNet

Does pruning by score behave, and does it scale?

PRUNING BY SCORE · 20% MISLABELED



Test accuracy after removing the lowest-valued (a, c) or highest-valued (b, d) points.

Compare accuracy after removing **low-valued** versus **high-valued** points.

IMAGENET · 1.28M IMAGES

Feature noise	Runtime	AUC
KAIROS	7m56s	0.869
SAVA	1h58m	0.817
Label noise	**Runtime**	**AUC**
KAIROS	7m52s	0.861
SAVA	1h58m	0.484

ResNet-50 features, one A100. AUC of the curve on the previous slide.

SAVA: a scalable, batched version of LAVA.
Kessler, Le & Nguyen, ICLR 2025.

> On ImageNet features, KAIROS scores 1.28M images in **about 8 minutes**, versus about 2 hours for SAVA, with higher reported detection AUC.

Zhu, Prashant, Cloninger, Salimi. KAIROS: Scalable Model-Agnostic Data Valuation. NeurIPS 2025.

LLM attribution: trace a specified completion loss

Grosse et al. · technical report · 2023

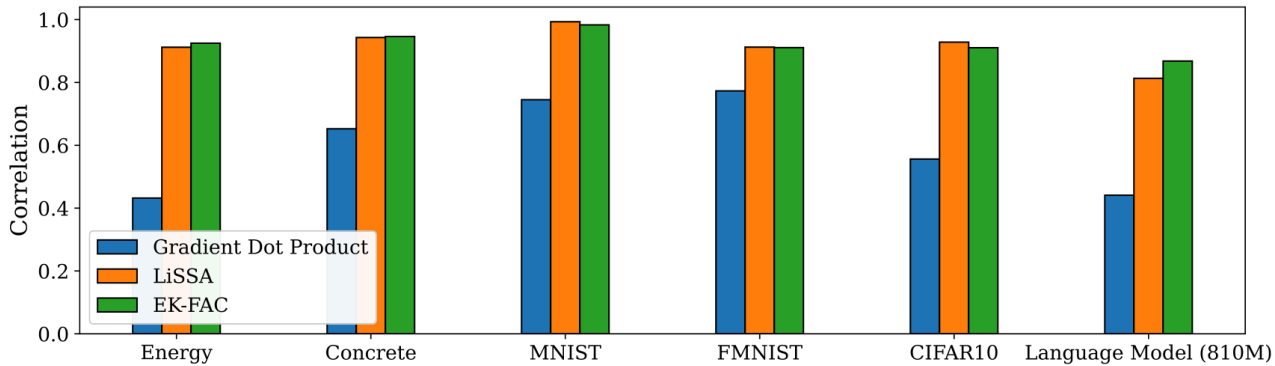
Studying Large Language Model Generalization with Influence Functions

Roger Grosse*†, Juhan Bae*†, Cem Anil*†

- Grosse and colleagues study training-example influence on language-model behavior.
- Query prefix and completion + candidate training sequences → gradients and approximate curvature.
- The output ranks local sensitivity of the specified completion loss.

LLM influence evidence: validate against the stated response target

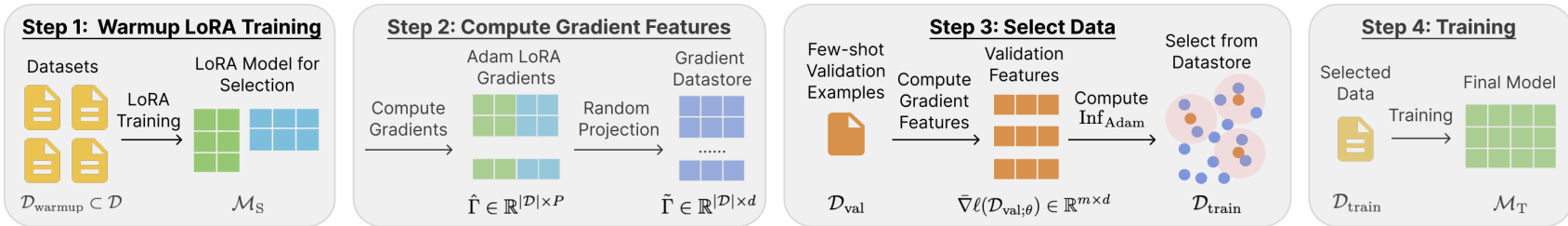
Grosse et al. · 2023 technical report



- The plotted target is proximal Bregman response, not exact leave-one-out retraining.
- Bars show Pearson correlation with PBRF; the language-model panel uses 810M parameters.
- A proximity-penalized response is useful when ordinary refitting is unstable.

LLM selection: LESS chooses instructions for a target capability

Xia, Malladi et al. · LESS · ICML 2024 · Original four-step pipeline



- Warm up a LoRA selection model on sampled candidate instructions.
- Build projected, Adam-aware candidate update features.
- Compare target-validation gradients with candidate features to select a budgeted subset.
- Train the target model on that subset; keep its test evaluation separate.

LESS evidence: selection helps some tasks, not every task

LESS · ICML 2024

	MMLU				TyDiQA				BBH			
Data percentage	Full (100%)	Rand. (5%)	LESS-T (5%)	LESS (5%)	Full (100%)	Rand. (5%)	LESS-T (5%)	LESS (5%)	Full (100%)	Rand. (5%)	LESS-T (5%)	LESS (5%)
LLAMA-2-7B	*51.6*	46.5 (0.5)	-	**50.2** (0.5)	*54.0*	52.7 (0.4)	-	**56.2** (0.7)	*43.2*	38.9 (0.5)	-	**41.5** (0.6)
LLAMA-2-13B	*54.5*	53.4 (0.1)	**54.6** (0.3)	54.0 (0.7)	*54.3*	53.0 (1.3)	**57.5** (0.8)	54.6 (0.3)	*50.8*	47.0 (1.6)	49.9 (0.5)	**50.6** (0.6)
MISTRAL-7B	*60.4*	60.0 (0.1)	60.6 (0.3)	**61.8** (0.4)	*57.7*	56.9 (0.2)	**61.7** (1.7)	60.3 (2.4)	*53.0*	54.5 (0.1)	**56.0** (0.8)	**56.0** (1.0)

- For Llama-2-7B, LESS at 5% gets 50.2% MMLU accuracy versus 51.6% for full data.
- TyDiQA gold-passage macro F1 is 56.2 versus 54.0 on a 0–100 scale.
- Parentheses report standard deviation across 3 training seeds.

Match the value to the data decision

- Influence estimates local response; Shapley allocates contribution to a defined game.
- KAIROS makes distributional discrepancy the objective and tests the resulting ranking.
- A useful ranking still needs verification, population checks and a justified intervention.